

```

1  n = int(input("Enter number of processes: "))
2  tq = int(input("Enter Time Quantum: "))
3  pid = []
4  at = []
5  bt = []
6  rt = []
7
8  for i in range(n):
9      pid.append(input(f"Enter PID for P{i+1}: "))
10     at.append(int(input("Arrival Time: ")))
11     bt.append(int(input("Burst Time: ")))
12     rt.append(bt[i]) # remaining time
13
14 ct = [0] * n
15 tat = [0] * n
16 wt = [0] * n
17
18 time = 0
19 queue = []
20 visited = [0] * n
21 completed = 0
22
23 while completed < n:
24     # Add arrived processes
25     for i in range(n):
26         if at[i] <= time and visited[i] == 0:
27             queue.append(i)
28             visited[i] = 1

```

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29     if len(queue) == 0:
30         time += 1
31         continue
32
33     i = queue.pop(0)
34
35     if rt[i] > tq:
36         time += tq
37         rt[i] -= tq
38     else:
39         time += rt[i]
40         rt[i] = 0
41         ct[i] = time
42         completed += 1
43
44     # Add new arrivals during execution
45     for j in range(n):
46         if at[j] <= time and visited[j] == 0:
47             queue.append(j)
48             visited[j] = 1
49
50     if rt[i] > 0:
51         queue.append(i)
52
53 for i in range(n):
54     tat[i] = ct[i] - at[i]
55     wt[i] = tat[i] - bt[i]
56
57 print("\nPID\tAT\tBT\tCT\tTAT\tWT")
58 for i in range(n):
59     print(f"{pid[i]}\t{at[i]}\t{bt[i]}\t{ct[i]}\t{tat[i]}\t{wt[i]}")

```

Enter number of processes: 3

Enter Time Quantum: 5

Enter PID for P1: A

Arrival Time: 0

Burst Time: 7

Enter PID for P2: B

Arrival Time: 2

Burst Time: 9

Enter PID for P3: C

Arrival Time: 1

Burst Time: 7

PID	AT	BT	CT	TAT	WT
A	0	7	17	17	10
B	2	9	21	19	10
C	1	7	23	22	15